

POWER BI PROJECT REPORT

SPOTIFY 2023 MUSIC INSIGHTS

THEORY - PROCEDURE - SCREENSHOTS

Name: Renu Lasya

UID: 24BAI70210

Section: 24NTTP-2A

Software: Microsoft Power BI Desktop

Dataset: Spotify 2023

Objective: To analyse Spotify 2023 song data in Power BI and create an interactive dashboard covering song popularity, audio features, platform presence, artist impact, playlist exposure and release trends.

1. INTRODUCTION / THEORY

Power BI is used to transform raw data into interactive reports and dashboards. For this project, the Spotify 2023 dataset is analysed using KPI cards, bar charts, column charts, line charts, scatter or bubble charts and tables. The analysis focuses on what the available data reveals about songs, artists, streaming activity, platform presence, playlists and release timing.

Concepts Used

1. Data preparation: checking fields, data types and creating useful derived fields.
2. Measures: creating Total Songs, Total Streams, Average Streams, Avg Energy and Avg Danceability.
3. Visual analysis: choosing charts according to the analytical question.
4. Formatting: applying consistent titles, labels, dark backgrounds and green accent colours.

General Procedure

1. Open Microsoft Power BI Desktop.
2. Import the Spotify 2023 dataset.
3. Open Transform data and prepare the dataset where required.
4. Create required measures and helper tables.
5. Create report pages and add suitable visuals.
6. Format and arrange the visuals consistently.
7. Review the final pages and save the Power BI project.

2. PAGE 1 - EXECUTIVE OVERVIEW

Theory: The first page provides a summary of the Spotify 2023 dataset using KPIs and overview visualisations.

Procedure:

Step 1: Create the report page and set a dark background.

Step 2: Add the title SPOTIFY 2023 MUSIC INSIGHTS.

Step 3: Create five Card visuals for Total Songs, Total Streams, Average Streams, Avg Energy and Avg Danceability.

Step 4: Create the Cross-Platform Presence chart using Spotify, Deezer, Apple Music and Shazam values.

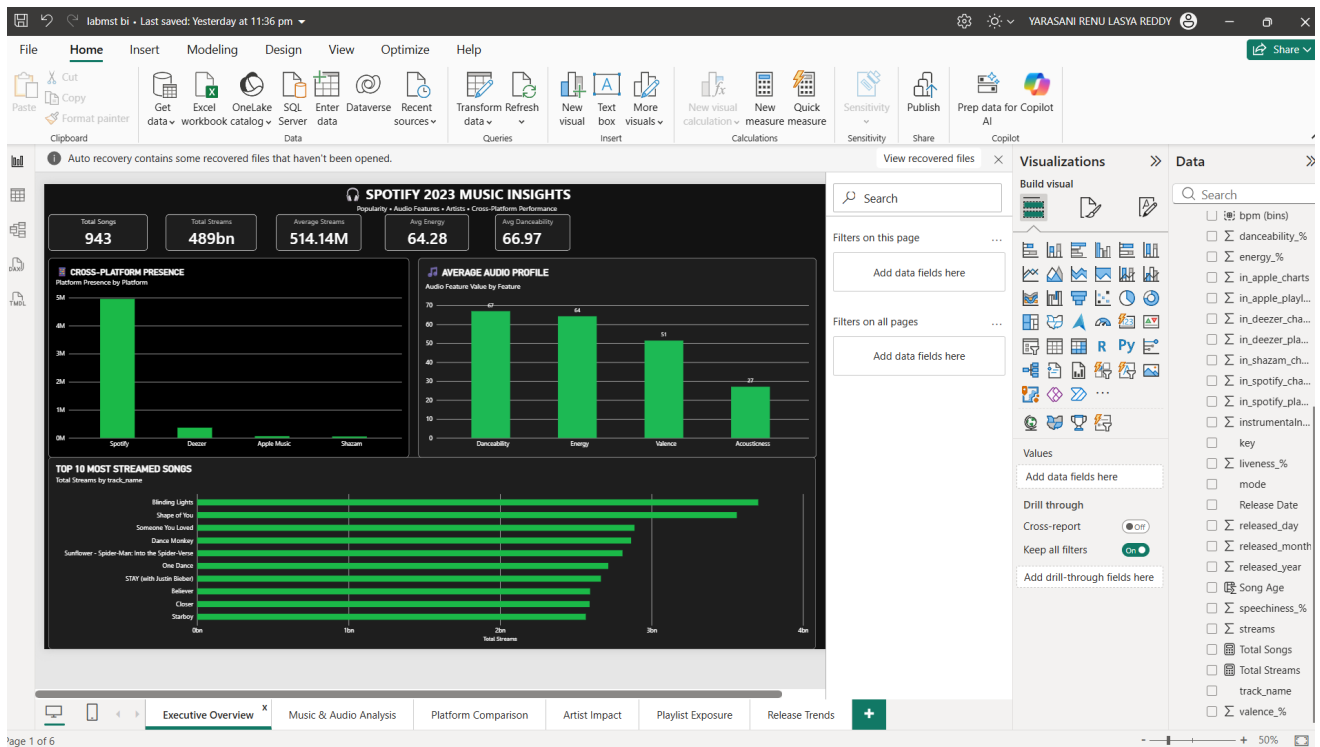
Step 5: Create the Average Audio Profile chart using Danceability, Energy, Valence and Acousticness.

Step 6: Create the Top 10 Most Streamed Songs bar chart using track_name and streams.

Step 7: Apply sorting, Top 10 filtering and consistent formatting.

Step 8: Align all visuals to form the Executive Overview.

Screenshot / Result:



3. PAGE 2 - MUSIC & AUDIO FEATURE ANALYSIS

Theory: This page analyses relationships and distributions in song audio characteristics.

Procedure:

Step 1: Create a new page named Music & Audio Analysis.

Step 2: Add the page title and subtitle.

Step 3: Create the Danceability vs Energy scatter or bubble chart.

Step 4: Put danceability_% on the X-axis and energy_% on the Y-axis.

Step 5: Use track_name as the detail or legend field and streams as bubble Size where required.

Step 6: Add artist, valence, BPM and acousticness fields to Tooltips.

Step 7: Create Song Tempo Distribution using bpm bins and count of tracks.

Step 8: Create Valence vs Streams.

Step 9: Create Acousticness Distribution.

Step 10: Arrange the four visuals in a 2 by 2 layout and format them consistently.

Screenshot / Result:



4. PAGE 3 - CROSS-PLATFORM PERFORMANCE

Theory: This page compares the presence and performance of songs across Spotify, Deezer, Apple Music and Shazam.

Procedure:

Step 1: Create a new page named Platform Comparison.

Step 2: Add the page title and subtitle.

Step 3: Create the Platform Presence comparison chart.

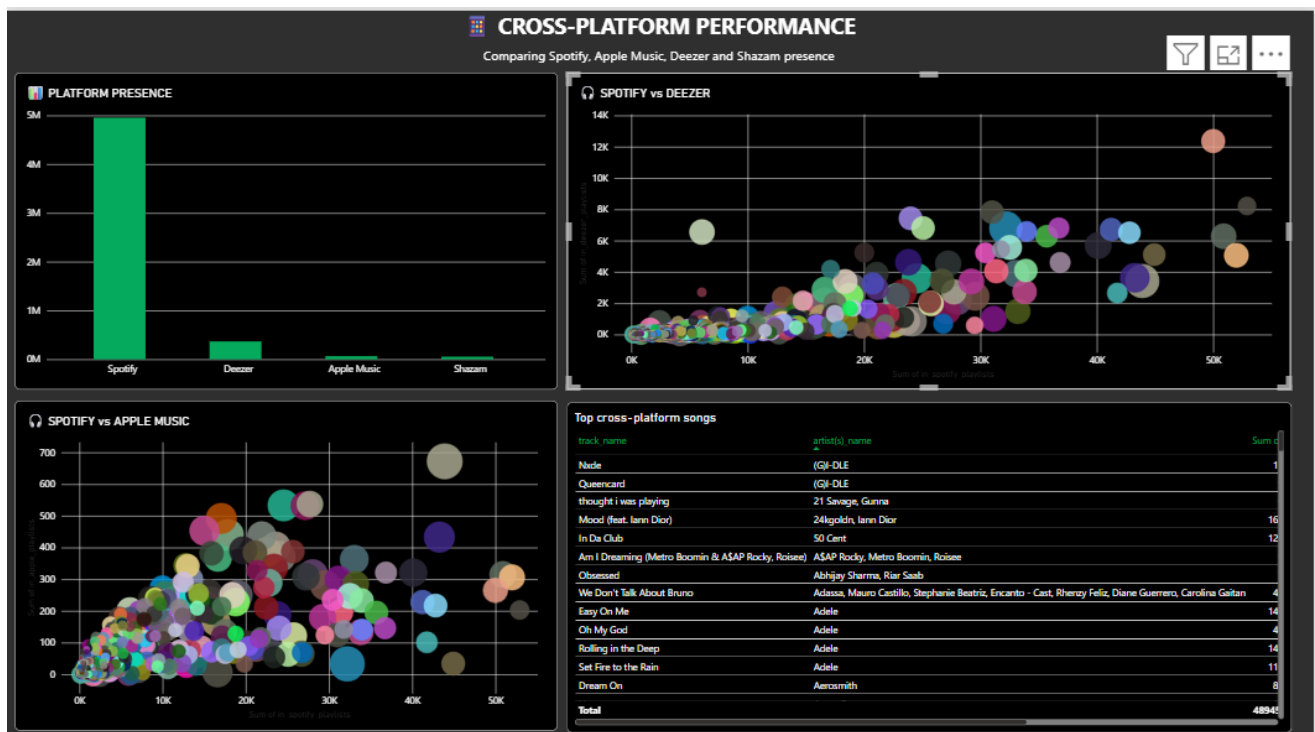
Step 4: Create the Spotify vs Deezer scatter or bubble comparison.

Step 5: Create the Spotify vs Apple Music comparison.

Step 6: Create the Top Cross-Platform Songs table with track_name, artist(s)_name and streams.

Step 7: Sort and format the table and align all visuals.

Screenshot / Result:



5. PAGE 4 - ARTIST IMPACT AND COLLABORATION

Theory: This page examines how artist involvement relates to song performance, playlist exposure and song counts.

Procedure:

Step 1: Create a new page named Artist Impact.

Step 2: Add the title ARTIST IMPACT & COLLABORATION.

Step 3: Create Artist Count vs Streams using artist_count on the X-axis and streams on the Y-axis.

Step 4: Use streams as bubble Size when required.

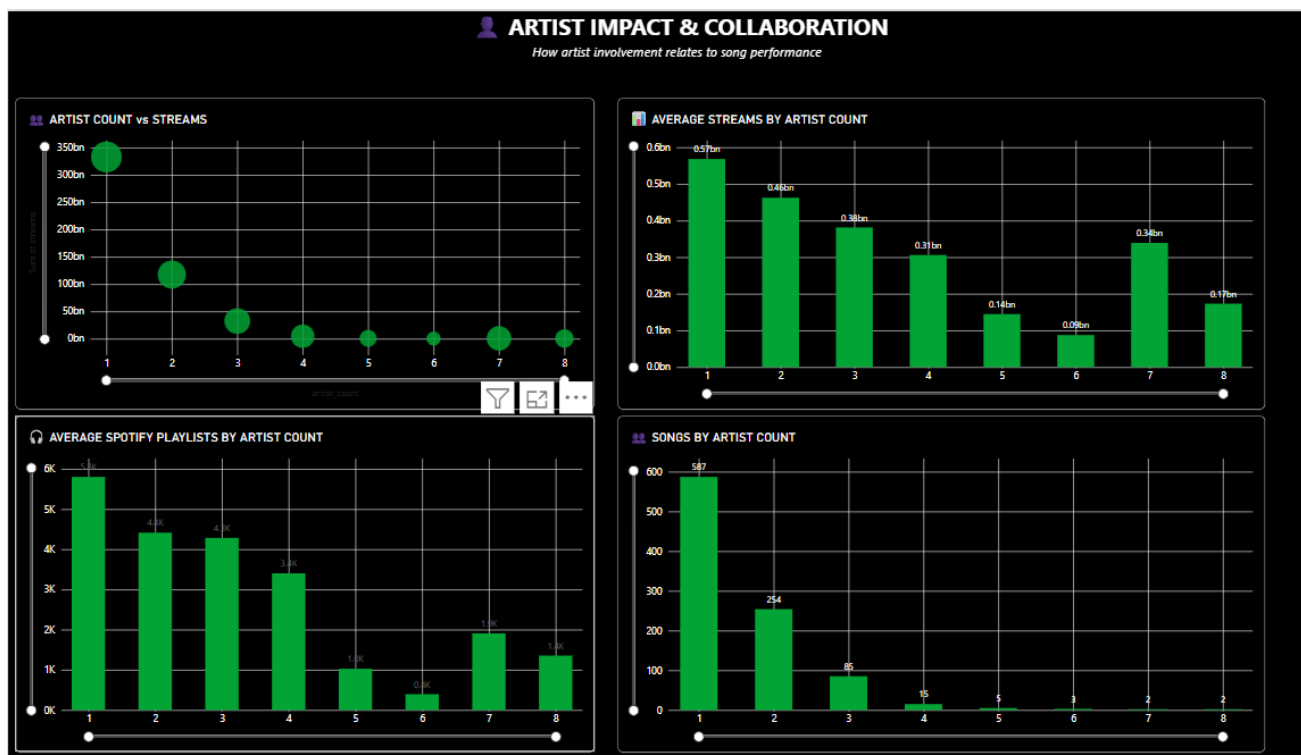
Step 5: Create Average Streams by Artist Count.

Step 6: Create Average Spotify Playlists by Artist Count.

Step 7: Create Songs by Artist Count using the count of track_name.

Step 8: Arrange the four visuals in a 2 by 2 layout and format them.

Screenshot / Result:



6. PAGE 5 - PLAYLIST EXPOSURE AND CHART PERFORMANCE

Theory: This page examines playlist exposure and its relationship with streaming performance.

Procedure:

Step 1: Create a new page named Playlist Exposure.

Step 2: Add the page title and subtitle.

Step 3: Create Spotify Playlists vs Streams using playlist exposure on the X-axis and streams on the Y-axis.

Step 4: Use song or track information for detail and streams for bubble Size where required.

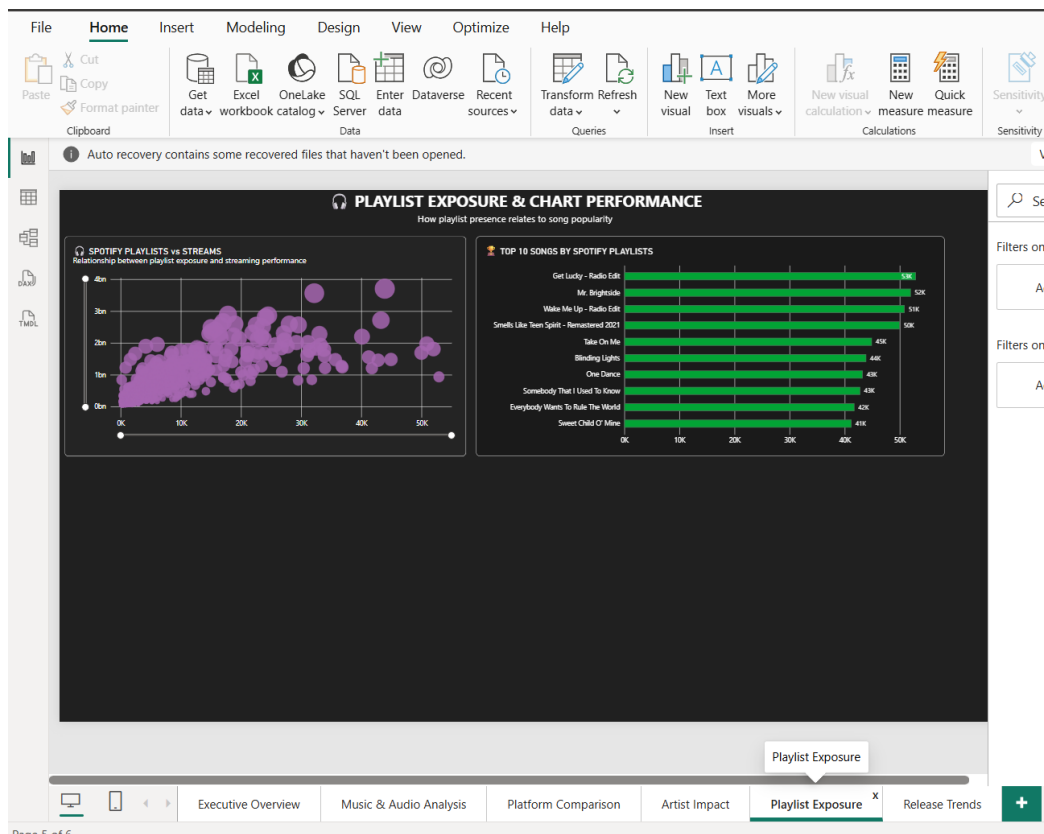
Step 5: Create Top 10 Songs by Spotify Playlists as a horizontal bar chart.

Step 6: Put track_name on the category axis and the Spotify playlist field as the value.

Step 7: Apply a Top 10 filter and sort in descending order.

Step 8: Format and arrange the visuals.

Screenshot / Result:



7. PAGE 6 - RELEASE TRENDS AND SONG PERFORMANCE

Theory: This page studies release year and release month patterns together with streaming performance.

Procedure:

Step 1: Create a new page named Release Trends.

Step 2: Add the title RELEASE TRENDS & SONG PERFORMANCE.

Step 3: Create Average Streams by Release Year using released_year and average streams.

Step 4: Create Song Releases by Year using released_year and count of track_name.

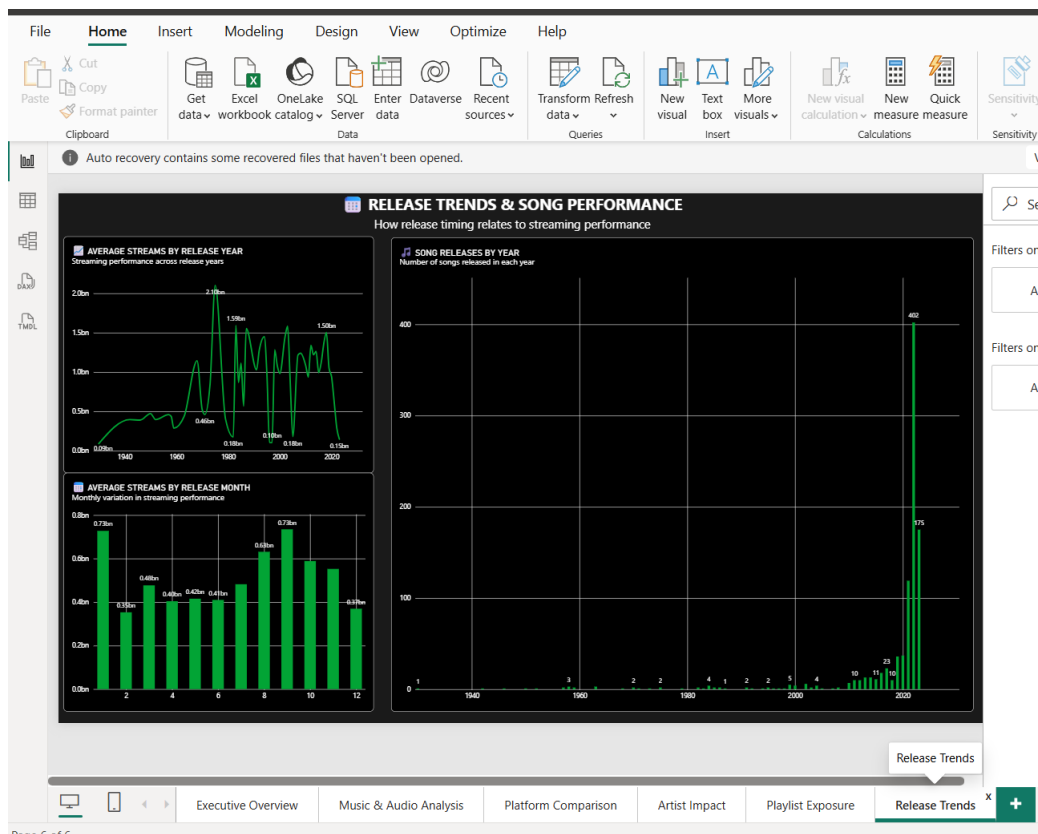
Step 5: Create Average Streams by Release Month using released_month and average streams.

Step 6: Add data labels where useful and format the visuals.

Step 7: If a Total Songs KPI is used, place the Total Songs measure in the Value field of a Card visual.

Step 8: Arrange the visuals and review the completed page.

Screenshot / Result:



8. CONCLUSION

The project was developed as a multi-page Power BI report for Spotify 2023 data. The final dashboard combines summary KPIs with music and audio analysis, platform comparison, artist impact, playlist exposure and release-trend analysis. The report is presented in a consistent dark theme with clear titles and visual organisation.

9. STUDENT DETAILS

Name: Renu Lasya

UID: 24BAI70210

Section: 24NTTP-2A